

FixFlow Viva Cheat Sheet

Simple answers for oral exam. Left = question. Middle = what to say. Right = exact place in the codebase / UI.

Tip: Section 9 lists landing/app animations: which UI piece, what motion, which file.

1. How Laravel works & project setup

Question	Simple answer (say this)	Where in codebase
How is Laravel structured?	Every request starts at one entry file. Then Laravel loads routes, checks, and pages.	public/index.php · bootstrap/app.php
How do we set up the project?	Install packages, copy .env, make app key, start Postgres, migrate + seed, run composer run dev.	composer.json · README.md · .env · .env.example
What are the main folders?	app = code, routes = URLs, resources/views = pages, database = tables and demo data.	app/ · routes/ · resources/ · database/
What is Artisan?	Laravel's CLI. We use it for migrate, seed, serve, queue, and Reverb.	README.md · php artisan ...
What is routing?	Routing means: this URL opens this page or runs this action.	routes/web.php · routes/channels.php
Which database?	PostgreSQL locally (pgsql). Tests still use SQLite in-memory.	config/database.php · .env (DB_CONNECTION=pgsql) · phpunit.xml

2. Blade (HTML pages)

Question	Simple answer (say this)	Where in codebase
What is {{ }} ?	It prints safe text on the page. Special HTML characters are escaped.	resources/views/layouts/app.blade.php
What is {!! !!} ?	It prints HTML as-is. We use it only for trusted icons.	resources/views/components/stat-card.blade.php
Blade comments?	Notes written as {{!-- ... --}}. Users never see them in the browser.	resources/views/welcome.blade.php
If / loops in Blade?	We use @if for conditions, @for / @foreach / @forelse for lists, and @while once in reports.	resources/views/repair-requests/* reports/index.blade.php
How do pages share one layout?	Guest pages use guest layout; app pages use app layout; landing uses marketing layout.	layouts/guest.blade.php · layouts/app.blade.php · layouts/marketing.blade.php
How does data reach the page?	The controller sends variables with the view. The Blade file prints those variables.	app/Http/Controllers/DashboardController.php · ReportController.php
What are @push and @stack?	A page can add extra JS/CSS. The layout prints those extras with @stack.	layouts/app.blade.php · invoices/create.blade.php
How do we make links?	We use route('name') or url('/') so links stay correct if URLs change.	routes/web.php · layouts/partials/sidebar-nav.blade.php

3. MVC, controllers, middleware

Question	Simple answer (say this)	Where in codebase
What is MVC here?	Model = database data. View = page. Controller = decides what happens for a request.	app/Models/RepairRequest.php · RepairRequestController.php · resources/views/repair-requests/
What does a controller do?	It checks the user, checks the form, saves or loads data, then shows a page or redirects.	app/Http/Controllers/RepairRequestController.php
What is middleware?	Code that runs before the page. Example: only admin can open admin pages.	app/Http/Middleware/EnsureUserHasRole.php · bootstrap/app.php
Middleware redirect / headers?	Unapproved technicians are redirected. Authenticated pages send Cache-Control: no-store (no back-button leak).	EnsureApprovedTechnician.php · AddFixFlowResponseHeaders.php · resources/js/session-guard.js
What is rate limiting?	We block too many tries in a short time (login, signup, ajax).	AppServiceProvider.php · routes/web.php (throttle:...)

4. Routes, forms, session, cookies

Question	Simple answer (say this)	Where in codebase
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Question	Simple answer (say this)	Where in codebase
Which HTTP methods do we use?	GET to open pages. POST to submit forms. PATCH/DELETE to update or remove (with @method).	routes/web.php · resources/views/invoices/*
What is CSRF?	A hidden token in forms so bad sites cannot submit forms for the user. We use @csrf.	resources/views/auth/login.blade.php · repair-requests/create.blade.php
How do we process a form?	Validate the input, then save it to the database, then redirect with a success message.	RepairRequestController@store · TechnicianApplicationController@store
What is a session?	Server memory of the logged-in user. After login we refresh the session. We also flash short messages.	routes/web.php (login / logout) · SESSION_* in .env
What are cookies here?	We save table density (compact/comfortable) in a cookie and read it on the layout.	PreferenceController.php · layouts/app.blade.php · top-navbar.blade.php

5. Database migrations & seeding

Question	Simple answer (say this)	Where in codebase
What is a migration?	A PHP file that creates or changes database tables. Easy to share and repeat.	database/migrations/
How do we design tables?	We define columns, unique fields, and links between tables in create migrations.	..._create_repair_requests_table.php · ..._create_invoices_table.php
How do we change a table later?	We add a new migration that only adds or removes columns (quotes, fulfillment, documents).	..._add_quote_fields... · ..._add_document_to_technician_applications · ...
Indexes and foreign keys?	Foreign keys link tables. Indexes/uniques make lookups faster and keep data unique.	..._create_messages_table.php · invoices migrations
What is seeding?	We fill the database with demo users, repairs, chat, invoices, and a waiting quote (RR-DEMO-QUOTE).	database/seeder/DatabaseSeeder.php
Rollback / refresh?	down() undoes a migration. For a clean demo we use migrate:fresh --seed.	README.md

6. Query Builder & Eloquent (database code)

Question	Simple answer (say this)	Where in codebase
What is Query Builder?	We write database queries in PHP (select, where, join) without raw SQL everywhere.	app/Http/Controllers/ReportController.php
What is Eloquent?	Each table has a Model class. RepairRequest model uses the repair_requests table.	app/Models/RepairRequest.php · Invoice.php · User.php
Insert / update / delete?	create to add, update to change, delete to remove. fillable lists safe fields.	Controllers + model \$fillable arrays
Where / select with Eloquent?	We filter lists and count rows on dashboards with where and count.	RepairRequestController · DashboardController
Joins and totals?	Reports join tables and use sum/count/groupBy. Charts use Chart.js on that data.	ReportController.php · resources/js/charts.js · reports/index.blade.php

7. Requests, JSON, Ajax

Question	Simple answer (say this)	Where in codebase
Do we have a REST API?	Not a full API project. Chat and notifications return JSON from normal web routes.	MessageController.php · NotificationController.php
Did we use Postman?	No Postman file in the project. Those same URLs can still be tested in Postman.	—
Different request types?	Forms send body data. Filters use URL params. Uploads send files. Fetch sends headers.	repair-requests/create · auth/technician-apply · resources/js/live-filter.js
File upload?	Customer: device image on repair form. Applicant: CV/certificate document (required).	RepairRequestController@store · TechnicianApplicationController@store · storage/app/public
Quote after diagnosis?	Technician sends quote → customer approves (repair continues) or declines (diagnosis fee only).	RepairRequestController quote/approve/decline · InvoiceService.php · repair-requests/show.blade.php

Question	Simple answer (say this)	Where in codebase
Redirect response?	After save we send the user to another page. Stripe sends them to Stripe's website to pay.	app/Http/Controllers/InvoicePaymentController.php
Ajax / Fetch?	JavaScript loads chat, filters, and notifications without a full page reload.	resources/js/chat.js · live-filter.js · notifications.js · unread-badges.js

8. Outside services (API)

Question	Simple answer (say this)	Where in codebase
Any external API?	Yes. Stripe for online invoice payment (test mode).	app/Services/StripePaymentService.php · InvoicePaymentController.php · .env STRIPE_*
Is chat an external API?	No. Live chat uses our own Laravel Reverb WebSocket server (local).	config/reverb.php · resources/js/chat.js · routes/channels.php · composer run dev

9. Frontend UI animations (what moves where)

Question	Simple answer (say this)	Where in codebase
Landing hero fade-in?	Hero title, CTAs, stats, and dashboard preview fade/slide up with staggered delays.	welcome.blade.php [data-hero-item] · app.css ff-fade-up · landing.js initHero()
Scroll reveal?	Sections fade in when they enter the viewport (features, services, steps, CTA).	welcome.blade.php data-reveal / data-reveal-child · landing.js initReveal() · app.css .ff-in-view
Number counters?	Hero stats count up from 0 when visible (eased animation).	welcome.blade.php data-counter · resources/js/landing.js initCounters()
Floating Live Chat / Invoice badges?	Soft up-down float loop on the hero mockup. Invoice badge starts later (delay).	welcome.blade.php .ff-float-badge · app.css @keyframes ff-float
Brand logo marquee?	Trusted brands strip scrolls horizontally forever; slows/pauses feel on hover.	welcome.blade.php .ff-marquee · app.css @keyframes ff-marquee
Reviews marquee?	Customer review cards scroll sideways; pause when the mouse is over them.	welcome.blade.php .ff-reviews-marquee · app.css @keyframes ff-reviews-marquee
Animated gradient text?	Some headings use a moving gradient color shift.	app.css .ff-gradient-text-animated · @keyframes ff-gradient-shift
Live green pulse on mockup?	Small "Live" dot on the fake dashboard pulses.	welcome.blade.php · app.css .ff-dashboard-live-dot · @keyframes ff-pulse-dot
Hover on cards / buttons?	Cards lift slightly; icons scale/rotate; primary buttons darken and press (active:scale).	app.css .ff-feature-card · .ff-btn-primary · welcome group-hover classes
Nav underline / spy?	Marketing nav highlights the section in view; underline grows with CSS transform.	landing.js initNavSpy() · app.css nav underline scale-x
Smooth scroll?	Clicking About/Services anchors scrolls smoothly (respects reduced-motion).	resources/js/landing.js initSmoothScroll()
Sidebar slide (app)?	On mobile, the admin/customer sidebar slides in/out with transform transition.	layouts/app.blade.php #sidebar · transition-transform duration-300
Chat typing indicator?	When the other user types, a "is typing..." line shows (Reverb whisper). Polling fallback if offline.	resources/js/chat.js · chat-panel component · Laravel Echo + Reverb
Reduced motion?	If the OS asks for less motion, we disable loops and show content instantly.	app.css @media (prefers-reduced-motion) · landing.js prefersReducedMotion()
Charts (not CSS animation)?	Reports/admin use Chart.js bar + doughnut charts (JS library, not keyframes).	resources/js/charts.js · reports/index.blade.php · dashboard/admin.blade.php

Short closing answer

FixFlow is a repair shop website built with Laravel + Blade + PostgreSQL. Users open Blade pages, controllers handle actions, and Eloquent saves data. We use migrations, seeders, middleware, forms with CSRF, sessions, file upload, quote approval after diagnosis, Chart.js reports, and Fetch/Reverb for live chat. Online payment uses Stripe. Landing motion is CSS keyframes + landing.js (no heavy animation library).